

PRANAV VERMA *Software Engineer*

pranavworksincode@gmail.com | +91 8920039114 | New Delhi, India | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Google-certified Cybersecurity Professional and Honors CS graduate specializing in secure automation, cloud-native infrastructure, and full-stack development. Experience spans Go/Kubernetes platform engineering, ML-driven threat defense, and secure e-commerce and EdTech architectures. Proficient in Python, full-stack JavaScript/TypeScript, SQL, and CI/CD pipelines.

EXPERIENCE

IT Consultant and Developer

Hoshiarpur Automobiles

04/2026 – Present

Hoshiarpur, Punjab, India

- Led diagnosis and resolution of critical network bottlenecks between the sales floor and service bays during peak hours, reducing downtime by 35%.
- Consulted with management to streamline data compliance and security protocols, enhancing data protection and reducing security breaches by 40%.
- Implemented CI/CD pipelines, improving deployment efficiency and reducing release downtime.

Jr. Software Engineer (PEP)

Energy and Mines | Énergie et des Mines

05/2022 – 08/2023

Guelph, Ontario, Canada

- Migrated fiscal platform to SharePoint Online, automating workflows and Power BI dashboards via Power Automate and Python, cutting manual processing time by 70%.
- Developed an award-winning business-technology mapping tool using Power Apps and Dataverse, earning 1st prize at the Ontario Public Services Hackathon, and built an AR web app showcased at the Foodland Ontario fair.

EDUCATION

Wilfrid Laurier University

Bachelor of Science in Computer Science (Hons.)

09/2019 – 10/2024

Waterloo, Ontario, Canada

- Graduated from a top-tier Canadian public (government) university; distinguished member of the university's Professional Experience Program (PEP), securing an internship at the Ministry of Northern Development and Mines/Energy and Mines.
- Founding President of the Artificial Intelligence Club, 2020–2022.

CERTIFICATES

Google Cybersecurity Professional Certificate — 8-course program covering Risk Management, Network Security, Assets/Threats/Vulnerabilities, Linux/SQL security automation with Python, and Incident Response (SIEM/IDS tools).

RECENT PROJECTS

Multi-Tenant Kubernetes Operator | *Go, Kubernetes, Kubebuilder, controller-runtime, Docker*

- Built a custom Kubernetes Operator (Go, kubebuilder, controller-runtime) automating the multi-tenant SaaS lifecycle — namespace provisioning, tier-based resource quotas, and dynamic TLS routing.
- Defined a custom Tenant CRD (saas.infra.io/v1alpha1) with LimitRanges to prevent noisy-neighbor resource contention.
- Engineered a level-triggered, idempotent reconciliation loop with zero-trust NetworkPolicies blocking cross-tenant traffic, and finalizers for zero-leak tenant teardown.

Binding Knowledge | *Next.js 14, React, TypeScript, Supabase/PostgreSQL*

- Built a 4-role EdTech platform (admin/teacher/student/parent) enforcing per-role access via Supabase Row-Level Security, with a Global Batch Manager for conflict-free student-batch assignment and automated 7-digit Roll ID generation.
- Shipped one-click PDF grade/attendance export (jsPDF), a multi-child parent portal, and real-time cross-role messaging with a leave-request workflow and live attendance sync.

Email Shield | *Python, ML/NLP, Heuristic Analysis*

- Built a multi-layered security tool using ML (NLP) and heuristic analysis to intercept phishing and malware in real time, engineering a real-time adaptive defense against evolving inbox threats.

NetSentinel | *Python, React, WebSockets*

- Built a full-stack network monitoring dashboard with raw packet capture and WebSockets for real-time, process-level telemetry visualization.

CK Cake Lounge | *Full-Stack Web, SQL*

- Built a secure full-stack e-commerce app with automated takeout ordering, payment processing, and real-time order tracking; hardened against SQL injection and XSS via input sanitization and parameterized queries.

SKILLS

Languages: Go, Python, TypeScript/Javascript, C++, Java, SQL, Prolog

Cloud & Web: Kubernetes, Kubebuilder, controller-runtime, Docker, CI/CD, Git, Next.js, React, Node.js, HTML, CSS

Databases & Security: Supabase/PostgreSQL, Dataverse, Linux, Splunk, Wireshark, tcpdump, SIEM/IDS

Other Tools: Power BI, Power Automate, MS/Google Suite, ChatGPT, Gemini